

Demonstrate Strings

Example demonstrating how to use **Java Strings**, showcasing common operations like finding length, changing case, checking contents, and joining text.

```
public class StringDemo {
    public static void main(String[] args) {
        // 1. Declare and initialize a String
        String text = " Hello, Java Learners! ";

        // 2. Get the length of the string
        System.out.println("Original length: " + text.length());

        // 3. Remove leading and trailing whitespace using trim()
        String cleanText = text.trim();
        System.out.println("Cleaned text: '" + cleanText + "'");

        // 4. Convert case (Uppercase / Lowercase)
        System.out.println("Uppercase: " + cleanText.toUpperCase());

        // 5. Extract a portion of the text using substring(startIndex, endIndex)
        // Extracts characters from index 7 up to (but not including) index 11
        String word = cleanText.substring(7, 11);
        System.out.println("Extracted word: " + word); // Outputs: Java

        // 6. Check if the string contains a specific word
        boolean hasJava = cleanText.contains("Java");
        System.out.println("Contains 'Java'? : " + hasJava);

        // 7. Replace characters or words
        String newText = cleanText.replace("Learners", "Experts");
        System.out.println("Replaced text: " + newText);

        String s1 = new String("Apple");
        String s2 = new String("Apple");
```

```
System.out.println(s1 == s2);    // ✗ Prints false (different memory objects)
System.out.println(s1.equals(s2)); // Prints true (same characters)
}
}
```

Expected Output

Original length: 25
Cleaned text: 'Hello, Java Learners!'
Uppercase: HELLO, JAVA LEARNERS!
Extracted word: Java
Contains 'Java'?: true
Replaced text: Hello, Java Experts!

Crucial Tip for Beginners: Comparing Strings

In Java, **never use ==** to compare the text content of two Strings. Always use the **.equals()** method.

- == checks if the strings are the exact same **object in memory**.
- .equals() checks if the **characters inside** the strings match.